Applied Machine Learning



Feature Engineering
Feature Transformations

Learning goals

- Normalization
- Box-Cox Transformation

FEATURE TRANSFORMATIONS I



- **Normalization:** The feature is transformed to have a mean of 0 and standard deviation of 1

$$z^{(i)} = \frac{x^{(i)} - \text{mean}(x)}{\text{sd}(x)}$$

Or use **robust** versions (median, IQR)

- **Box-Cox Transformation:** Stabilizes variance, makes the data more normal distribution-like

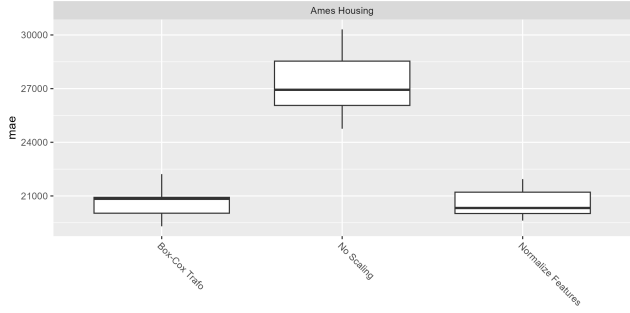
$$z^{(i)} = \begin{cases} \frac{\left(x^{(i)}\right)^{\lambda} - 1}{\lambda} & \text{if } \lambda \neq 0 \\ \log\left(x^{(i)}\right) & \text{if } \lambda = 0 \end{cases}$$

Note: Box-Cox can also be applied to the target – see separate slide set.

FEATURE TRANSFORMATIONS I



To illustrate the effect of transforming the features we evaluate a k-NN learner without scaling, with normalization, and with a Box-Cox transformation:



OTHER COMMON TRANSFORMATIONS I



- Polynomials: $x_j \longrightarrow x_j, x_j^2, x_j^3, ...$
- Interactions: $x_j, x_k \longrightarrow x_j, x_k, x_j \times x_k$
- Basis expansions: BSplines, TPB, ...
- Fourier expansions

These transformations are used to improve simple models, e.g. linear regression, and most likely will **not** improve complex machine learning models.

FEATURE TRANSFORMATIONS - OTHER DATA TYPES I



Feature transformations allow handling a variety of data types:

- **Dates:**
  - Time since X
  - Birthday $\rightarrow$ age
  - Extract month, day of the week, ...
- **Other:**
  - Use outputs of neural networks (images, text)
  - Bag-of-words (text)
  - Statistics (time-series)

FEATURE TRANSFORMATIONS - SUMMARY I



- Transformations of the target variable can make modelling highly skewed data easier (see separate slide set on target transformations)
- Scaling single features can lead to better results
- For some learners, e.g. tree-based methods scaling has no effect!
- Feature transformations can further improve models - use domain knowledge!